

arXiv Radar

Daily digest of arXiv papers scored for relevance by Claude (via Bedrock), delivered via SES.

Project layout

```
arxiv-radar/  
├── lambda/          # Lambda handler and library code  
├── config/          # config.json - uploaded to S3 on deploy  
├── infra/           # CDK stack  
├── tests/  
│   └── integration/
```

Prerequisites

1. SES sender address verified in your AWS account
2. Bedrock model access enabled for `anthropic.claude-haiku-4-5-20251001-v1:0` in your region
3. CDK bootstrapped in target account/region

Deploy

CDK picks up your account and region automatically from your AWS credentials/profile.

```
cd infra && ./deploy.sh
```

Test locally

```
cd lambda  
uv pip install -r requirements.txt  
  
export SES_SENDER=mjdinsmore@gmail.com  
export SES_RECIPIENT=mjdinsmore@gmail.com  
export CONFIG_ENV=dev          # reads s3://arxiv-radar-dev/config.json  
  
python -c "import handler; print(handler.handler({}, None))"
```

Bedrock and S3 auth use your local AWS credentials — no API key needed.

Run unit tests

```
uv run pytest
```

Run integration test (live arXiv fetch)

```
RUN_INTEGRATION_TESTS=1 uv run pytest tests/integration -s
```

Tuning

All tuneable settings live in `config/config.json`. Edit the file and re-run `./infra/deploy.sh` — it uploads the config to S3 automatically after every deploy. No Lambda redeployment is needed for config-only changes; just run:

```
aws s3 cp config/config.json s3://arxiv-radar-dev/config.json
```

Key fields:

Field	Default	Description
<code>categories</code>	<code>["q-bio.GN", ...]</code>	arXiv category codes. Full list: https://arxiv.org/category_taxonomy
<code>min_score</code>	<code>6.0</code>	Papers below this score are excluded from the digest
<code>top_n</code>	<code>5</code>	Max papers included regardless of score
<code>research_interests</code>	<i>(your interests)</i>	Most important tuning lever — be specific
<code>scoring_system</code>	<i>(prompt template)</i>	Full Claude prompt; use <code>[RESEARCH_INTERESTS]</code> as placeholder

Useful category codes for biotech/ML: — `cs.AI` artificial intelligence — `q-bio.GN` genomics — `q-bio.QM` quantitative methods — `q-bio.TO` tissues and organs, tumor growth

Schedule

EventBridge cron is `30 6 * * ? *` (06:30 UTC). Edit in `infra/app.py` to match your timezone. arXiv posts new submissions around 00:00 UTC, so 06:00–08:00 UTC is a good window.

arXiv rate limit retries

If arXiv returns HTTP 429, the Lambda schedules a retry via EventBridge Scheduler 13 minutes later, passing the attempt count in the event payload. It retries up to 18 times (~4 hours total) before giving up. The 13-minute interval is intentionally odd to avoid re-syncing with other clients polling on regular intervals.

Cost estimate (rough)

Resource	Est. monthly
Lambda (5 min/day, 512 MB)	~\$0.01
Claude Haiku via Bedrock (80 abstracts/day, ~600 tokens each)	~\$0.40
SES (1 email/day)	~\$0.00
S3 (1 tiny config file, ~30 GETs/month)	~\$0.00
Total	~\$0.40/month